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MILITARY SPECIFICATION

GRINDING MACHINES, TOOL AND CUTTER, FLUTE AND RADIUS, COMPUTER NUMERICALLY CONTROLLED (CNC)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers computer numerically controlled (CNC) tool and cutter grinders with equipment and accessories specified herein.

1.2 Classification. This specification covers grinders of the following types, styles, and sizes. The type, style, and size to be furnished shall be as specified (see 6.2.1).

Type I - Conventional workhead/tailstock workholding assembly

Style A - Worktable/swiveling sub-table arrangement

Size	Maximum distance between centers (inches)	Number of NC axes
1	23-1/2	2, 3 or 4
2	33	2, 3 or 4

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5051, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Style B - Single, nonswiveling worktable

Size	Maximum distance between centers (inches)	Number of NC axes
1	17	3 or 4
2	35	3, 4, 5 or 6

Type II - Rotary table/traversing worktable arrangement

Size	Maximum tool length (inches)	Number of NC axes
1	12	5
2	16	7

2. APPLICABLE REFERENCE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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U. S. DEPARTMENT OF LABOR

29 CFR 1910 - Occupational Safety and Health Standard (OSHA)

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this specification as specified. Unless otherwise specified, DoD adopted documents shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issue of documents not listed in the DoDISS shall be the issue of the non-Government documents which are current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA MG 1 - Motors and Generators.

ANSI/NFPA 70 - National Electrical Code.

ANSI/NFPA 79 - Electrical Standards for Industrial Machinery.

ANSI B5.10 - Machine Tapers.

ANSI B7.1 - Safety Requirements for the Use, Care, and Protection of Abrasive Wheels.

ANSI B11.9 - Safety Requirements for the Construction, Care and Use of Grinding Machines.

ANSI B74.2 - Specifications for Shapes and Sizes of Grinding Wheels, and Shapes, Sizes, and Identification of Mounted Wheels.

ANSI B74.13 - Marking for Identifying Grinding Wheels and Other Bonded Abrasives.

ANSI B94.19 - Milling Cutters and End Mills.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3351.)

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103-1187.)

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machine Tool Gearing.

AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears.

(Application for copies should be addressed to the American Gear Manufacturer's Association, Standards Department, Suite 1000, 1901 North Fort Meyer Drive, Arlington, VA 22209-1695.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO 54 - Cylindrical Gears for General Engineering and for Heavy Engineering, Modules and Diametral Pitches of.

(Application for copies should be addressed to the International Organization for Standardization, c/o the American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018-3351.)

ELECTRONIC INDUSTRIES ASSOCIATION (EIA)

RS-227 - One Inch Perforated Tape.

RS-232 - Interface between Data Terminal Equipment and Data Communication Equipment Employing Serial Binary Data Interchange.

RS-244 - Character Code for Numerical Machine Tool Control Perforated Tape.

RS-267 - Axis and Motion Nomenclature for Numerically Controlled Machine Tools.

RS-274 - Interchangeable Variable Block Format for Positioning, Contouring, and Contouring/Positioning Numerically Controlled Machines.

RS-281 - Construction Standards - Numerical Machine Tool Control.

RS-358 - Subset of American National Standard Code for Information Interchange for Numerical Machine Control Perforated Tape.

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(Application for copies should be addressed to Electronic Industries Association, Engineering Department, 2001 Eye Street, N.W., Washington, DC 20006-5009.)

(Non-Government standards and other publications are normally available from the organizations which prepare or distribute the documents. These documents also may be available in or through libraries or other information services.)

2.3 Order of precedence. If text of this specification conflicts with the references cited (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

* 3.1 Design. The cutter grinder and its components shall be new, of the manufacturer's latest numerically controlled design, and shall meet the requirements of this specification. The machine shall be an electrically powered, motor driven unit with a solid-state, soft-wired, integrated circuit numerical control sytem. It shall automatically grind all configurations of end mills as specified in ANSI Standard B94.19 from 1/4 inch through at least 2-1/2 inches in diameter as well as odd-number fluted end mills. The machine shall grind the primary and secondary clearance angles including radial relief on the periphery, on the corner radius including full ball, and on the end of the cutter, the face of the flutes (hook angle) along the axis of the cutter, and the end gashes of all types of end mills. It shall also grind and resharpen reamers, counterbores, hobs, and other metal-cutting tools. All functions shall be performed in one set-up and with one wheel. The design shall provide convenient, efficient, and safe operation. The machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair.

3.1.1 Measurement systems. Unless otherwise specified (see 6.2.1), either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in US units. These measurements may be converted to SI units through the use of the conversion factors and methods specified in FED-STD-376.

3.1.1.1. Measuring and indicating device calibrations. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used to graduate measuring and indicating devices such as scales, depth stops, carriage stops, dial indicators, pressure gauges, temperature indicators, and other similar devices. When only one system of graduation is acceptable, the particular graduation

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required shall be as specified (see 6.2.1). Regardless of the measurement system used, all measuring and indicating devices on the machine shall be graduated in the same system. All feed dials shall have independent zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.1.2 Reclaimed materials. The machine may contain reclaimed materials if those materials will not jeopardize the machine's intended use and performance. The reclaimed materials shall have been reprocessed, remanufactured, or recycled to restore them to the same chemical composition and physical properties as the materials originally selected for use on the machine.

* 3.1.3 Energy efficiency. The machine shall achieve the highest degree of energy efficiency as governed by the developments available within the industry.

3.1.4 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety devices shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with all of the requirements of ANSI B11.9 and 29 CFR 1910 that are applicable to the machine itself. When the requirements of the ANSI and 29 CFR 1910 standards conflict, the 29 CFR 1910 standards shall take precedence. Additional safety and health requirements shall be as specified (see 6.2.1).

3.1.5 Mercury restriction. The machine shall not contain mercury nor be exposed to free mercury during manufacture.

3.1.6 Asbestos restriction. Asbestos shall not be used on or in the machine.

3.1.7 Environmental protection. Under the operating, service, transportation, and storage conditions described herein, the machine shall not emit materials hazardous to the ecological system as prescribed by Federal, state, or local statutes in effect at point of installation.

3.1.8 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances.

3.2 Construction. The machine shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand

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all forces encountered during operation of the machine at its maximum rating and capacity without distortion.

3.2.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.2.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.2.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed to prevent loosening. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.2.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.2.5 Painting. Unless otherwise specified (see 6.2.1), the machine shall be painted in accordance with the manufacturer's commercial practice.

3.2.6 Threads. All threaded parts used on the machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

* 3.2.7 Dials. All rotating dials or handwheels which are used to position machine axes or for tool movement shall be graduated in increments not larger than 0.0001 inch. Dial and handwheel circumferences shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position.

* 3.2.8 Gears. All gears and pinions of spindle and axis drive trains shall be designed and manufactured of a suitable ferrous material to meet or exceed the requirements of AGMA 360.02 and 390.03 for the US system or ISO 54 for the SI system. The gears shall be of proper width and size to transmit full-rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train shall be hardened and ground steel. Hardness shall be not less than Rockwell C-45.

* 3.2.9 Machine ways. All machine ways shall be protected from grinding dust, dirt, coolant, and other hazards that may cause damage or result in

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inaccurate functioning of any members. The machine ways shall be hardened to Rockwell C-40 and ground to 30 microinch and suitable for the precision numerically controlled machines covered by this specification. Linear slide movement over the ways shall be supported by roller packs, non-metallic liners, or anti-friction ways.

3.2.10 Electromagnetic interference control. When specified (see 6.2.1), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystems class and the emission and susceptibility requirements shall be as specified.

3.3 Components. The machine shall consist of the following basic components. Any additional components necessary for the operation of the machine shall be furnished in accordance with manufacturer's standard practice.

3.3.1 Bed. The bed shall be a floor-mounted structure of iron or steel, ribbed and braced. Means shall be provided to support the wheelhead assembly, the workhead table, and workhead assembly. When applicable, coolant reservoirs and dust extractors may be housed in the base or provided as separate external units. An access door(s) or removable plate(s) shall be provided as necessary for servicing components housed within the base.

* 3.3.2 Workhead table. The workhead table shall be a one-piece casting or weldment. On type I machines, the table shall be mounted on guideways provided on the bed and arranged to traverse longitudinally along the bed. On type I, style A machines, the worktable shall carry a swiveling sub-table to support the workhead assembly and tailstock (see 3.3.2.1). On type I, style B machines, the workhead table shall directly support a sliding workhead assembly and tailstock. On type II machines, the workhead table shall be a component in a rotary table assembly (see 3.3.2.3). Means shall be provided to move the table manually for set-up. The length of table stroke shall be infinitely variable within the total grinding stroke of the machine. On type I, style B machines, the movement of the sliding workhead assembly and tailstock along the worktable shall be guided by the double vee cross-section and T-slot configuration of the worktable. The surface of the worktable shall be machine finished for a work mounting surface. When applicable, coolant troughs shall be provided to drain spent coolant into a reservoir. Means shall be provided to prevent table movement after positioning. Stops shall be provided at each end of travel to prevent table overtravel. Workhead table travel, size, swivel, and feed rates, as applicable, shall meet the requirements of table I.

3.3.2.1 Swiveling sub-table. On type I, style A machines, the workhead table shall support a sub-table which swivels and can be locked in any position. The sub-table surface shall be machine finished for a work mounting surface. The sub-table shall have a double vee cross-section and T-slot configuration. The sub-table T-slot shall guide the movement of the sliding workhead assembly and tailstock movement. When applicable, means shall be provided to drain spent coolant into its reservoir.

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3.3.2.2 Tailstock. Type I machines shall be provided with a tailstock. The tailstock shall be adjustable along the table in alignment with the workhead when the workhead is set on zero. The tailstock taper, or other internal configuration, shall be as specified (see 6.2.1). The tailstock spindle shall be designed to allow tension adjustment with a clamping device provided to lock the spindle with desired tension.

* 3.3.2.3 Workhead rotary table assembly. On type II machines, a base slide mounted on the base guideways shall support a workhead rotary table and shall allow the rotary table to be positioned longitudinally. The rotary table shall be of the contouring or positioning type, as specified (see 6.2.1). The workholder shall be mounted on a sub-table assembly consisting of a table and saddle unit mounted on top of the rotary table. These two linear axes, the table and the saddle, shall permit positioning the tool in relation to the center of rotation of the rotary table. The table shall travel parallel to the longitudinal base slide and the saddle shall travel perpendicular to the table travel. The rotary table, the table, and the saddle, as applicable, shall be integral components of the machine. Traversing axes shall be provided with stops to prevent overtravel. Workhead rotary table assembly linear, rotary, and swivel movement shall meet the requirements of table I.

* 3.3.3 Workhead assembly. The workhead assembly shall consist of a housing, spindle assembly, drive motor, and all other components necessary to perform all required functions. The workhead assembly shall be mounted on the workhead swiveling sub-table on type I, style A machines; on the workhead table on type I, style B machines; and on the workhead rotary table assembly on type II machines. The workhead assembly shall allow the cutter to tilt and index about its own axis. Cutter rotation shall be adjustable to an angle equal to the helix angle of the cutter so that a gash can be made to center with negative, positive, or radial rake. Means shall be provided to secure the desired rake angle of the cutter after positioning. The workhead assembly on type I machines and the workhead rotary table assembly on type II machines shall permit swiveling of the workhead in a horizontal plane. Stops shall be provided to limit the swivel in both directions. Longitudinal positioning of the workhead shall be provided on type I machines by sliding in the table T-slots; on type II machines, by movement of the table and base slide, with cross-travel provided by the saddle of the rotary table and sub-table assembly. The workhead assembly shall provide clearance for the grinding wheel(s) to permit grinding to the nose of the workholding device. The spindle shall accept quick change type adapters used in the workholder. Adapters shall be of the collet type designed to accept precision ground round collets for maximum accuracy. The workholder shall index 2, 3, 4, 6, or 8 flute mills. The workholding spindle shall be mounted on antifriction bearings and shall accept adapters for special collets sizes when specified (see 6.2.1). Tapers shall conform to ANSI B5.10.

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* 3.3.4 Wheelhead column assembly. The wheelhead column assembly shall consist of a vertically adjustable column and cross slide. The column shall allow vertical wheelhead assembly travel and the cross slide shall allow cross axis wheelhead assembly travel perpendicular to the workhead table longitudinal travel. The wheelhead assembly shall be mounted on either the column or cross slide, as applicable. The column shall be equipped with means to prevent the entry of grit and dust. Means of preventing movement of both the column and cross slide after positioning shall be provided.

* 3.3.5 Wheelhead assembly. The wheelhead assembly shall consist of a housing, a grinding wheel spindle, and motor. The wheelhead assembly shall mount on the the wheelhead column assembly. On type I machines, the base shall allow the wheelhead to rotate about its vertical axis and the spindle housing shall allow the wheelhead to tilt above and below center. The wheelhead shall have at least 4 speeds or, when specified (see 6.2.1), shall be infinitely variable throughout its entire speed range. All belts required to provide necessary wheel speeds shall be furnished. The wheelhead shall be designed for using straight, cup or dish wheels of diameters which shall permit use of the full capacity of the grinding machine. The spindle nose shall accept taper style precision wheel mounts which, once mounted and dressed or indicated in, can be dismounted and remounted without loss of trueness, balance, or location along spindle axis. Each grinding wheel spindle shall be mounted on preloaded ball bearings, sealed and permanently lubricated to provide vibration-free operation at all spindle speeds. Means shall be provided to prevent movement of the wheelhead unit after positioning. Wheelhead travel, swivel, tilt, spindle speeds, and horsepower shall meet the requirements given in table I.

3.3.6 Wheel dressing and truing. Means shall be provided to dress and true the grinding wheel with either one industrial diamond or a cluster diamond, as specified (see 6.2.1). When specified (see 6.2.1), the machine shall have programmable numerically controlled wheel truing and dressing. Appropriate dress rates shall be established by the control based on desired wheel speed and dress load. Dress modes shall be selective from sequence to sequence. The wheel dresser shall be manufacturer's standard, readily adjustable for various work sizes, and shall include means for fine adjustment. Provisions shall be made to compensate for change of grinding wheel size with wear so the size and geometry of the cutter being ground with not be changed.

3.3.7 Grinding wheels. Grinding wheels shall conform to ANSI B74.2. The wheels shall be marked in accordance with ANSI B74.13 and shall meet the safety requirements established in ANSI B7.1. All grinding wheels shall be properly guarded.

* 3.3.8 Locator fixture. When specified (see 6.2.1), a locator fixture shall be provided so that the cutter to be ground can be preset in a toolholder for length and radial position. Precision toolholders shall be

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provided for all shank sizes applicable for the machine type, style, and size specified. The fixture shall accommodate right- and left-hand cutters.

3.3.9 Lubrication. An automatic lubrication system shall be provided for all parts and mechanisms requiring a continuous flow of lubrication. Means shall be provided to determine oil level in each reservoir. Mechanisms requiring periodic manual lubrication shall be readily accessible for servicing. Each recirculating system shall include a cleanable or replaceable filter.

* 3.3.10 Coolant system. Unless otherwise specified (see 6.2.1), a coolant system shall be provided. The coolant system shall include a sump, or reservoir, a power driven pump, and piping. The coolant reservoir shall be of sufficient capacity to permit full-flow cooling. The coolant system shall include a baffle(s) or strainer, easily removable for cleaning. The reservoir shall be equipped with means for emptying and cleaning. Means shall be provided to permit the operator to direct and control the amount of coolant flow. The coolant system shall be numerically controlled.

3.3.11 Dust extractor. When specified (see 6.2.1), a dust extractor and all necessary connections shall be provided to be utilized during dry grinding. The dust extractor shall be connected to the machine's electrical system with start/stop pushbuttons conveniently located on the control panel.

3.3.12 Electrical system. Unless otherwise specified (see 6.2.1), the electrical system shall conform to ANSI/NFPA 79. Each machine shall draw all of its electrical power from a single 230/460-volt, 3-phase, 60-hertz (Hz) circuit with a maximum line voltage fluctuation of ±10% of the rated voltage. Each machine shall have a fused safety disconnect switch or circuit breaker. The machine shall be initially wired for operation on 460 volts. An identified terminal for grounding the machine when it is installed shall be mounted in or near the disconnect switch. The terminal shall be suitable for connecting the size grounding conductor specified in ANSI/NFPA 79 for the disconnect fuse rating.

3.3.12.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Alternating current (AC) motors shall be designed to operate on 60-Hz. Unless otherwise specified (see 6.2.1), each motor enclosure shall meet the requirements for a dripproof enclosure. All motors shall meet the requirements of ANSI/NEMA MG-1 and ANSI/NFPA 70. Separate motors shall drive each powered axis, the grinding wheel, work spindle, coolant system, and dust extractor, as applicable.

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3.3.12.2 Main electrical disconnect. A manually operated fusible disconnect switch or circuit breaker of adequate size to carry the entire electrical load of the equipment shall be provided for connection to the primary power source specified.

* 3.3.12.3 Control circuit voltage. Main and auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having a secondary voltage not exceeding 115 volts.

* 3.3.13 Numerical control. The machine offered shall be completely programmable through a microprocessor type computerized numerical control (CNC) as described in 3.3.13 through 3.3.13.35. The machine control unit (MCU) shall provide automatic control of machine functions, operating modes, axes positioning and feed motion, wheelhead speed and direction of rotation, rapid traverse rates, axis dwell times, dress type, dress rates, dress frequency (when applicable), and shall initiate action should a machine fault occur. The axes to be numerically controlled shall be as specified (see 6.2.1). The axes shall be identified in accordance with EIA Standard RS-267. The control shall have linear and circular interpolation and be capable of up to seven axes simultaneous control, as applicable. Control features shall include a programmable interface, buffer storage, fixed cycles, part program storage, part program and buffer edit capability, and control diagnostics. The MCU shall direct machine functions from part program data stored in memory, through manual data input (MDI), or from data directly input from the input media specified in 3.3.13.1. All necessary executive diagnostic and interface program routines shall be furnished in the form of one-inch, eight-channel binary coded decimal punched tape or other media as appropriate for input media specified (see 6.2.1) for controls having volatile memory. The control unit shall conform to EIA Standard RS-281. A line voltage of $\pm$10 percent from normal shall not adversely affect the numerical control function. The control shall be capable of functioning in ambient temperatures ranging from 50 to 120° F and relative humidities ranging from 5 to 95 percent noncondensing. The control shall automatically shut down when internal operating temperature exceeds safe operating temperature. Control resolution shall be 0.0002 inch for each linear axis. The system resolution shall be 0.01 degrees for each rotary axis.

* 3.3.13.1 Data input. Unless otherwise specified (see 6.2.1), the input media for the control shall be one-inch, eight-channel punched tape meeting the requirements given in EIA Standard RS-227. The data input format and codes shall comply with EIA Standards RS-274 and RS-358 or RS-244, as specified (see 6.2.1). The control shall be equipped with a photoelectric type reader capable of holding, feeding and reading tape prepared in the format specified at a rate not less than 120 characters per second. Unless otherwise specified (see 6.2.1), the tape reader shall be supplied with standard feed and take-up spoolers, 5-1/4 inches in diameter. The tape reader shall be housed in a dust free area of the numerical control cabinet, easily accessible for cleaning and maintenance. The control shall have a tape error detection capability to continuously check for the following errors:

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a. Misaligned sprocket holes.
b. No sprocket holes.
c. Improper parity.
d. Overshoot.

Any of the above errors shall halt machine functions and illuminate an error light or a message on the data display screen to indicate a fault condition.

* 3.3.13.1.1 Manual data input. Unless otherwise specified (see 6.2.1), a manual data input (MDI) unit shall be furnished to permit manual entry of alphabetical, numerical and special characters necessary for control of all programmable machine functions. The manually input data shall be immediately displayed on the data display to permit data verification. The MDI shall be capable of entering a block of data into memory upon command.

* 3.3.13.2 Data display. Unless otherwise specified (see 6.2.1), all operating data shall be displayed on a cathode ray tube (CRT). The system shall display all operating data, actual and command, in alpha-numeric form from memory storage, buffer storage, and active data. Display data shall include slide position data, auxiliary functions, and tape and MDI input, as applicable. CRT display capacity shall be at least 256 characters (8 lines, 32 characters per line).

3.3.13.3 Control system modes of operation. The operational modes of the control shall include manual, automatic, single block, MDI, sequence number search, and optional stop.

3.3.13.4 Departure control. The maximum departure permitted by a single command in either incremental or absolute mode shall be not less than plus or minus 99.9999 inches. The control shall automatically control the acceleration and deceleration of the slides. The resolution of slide movement in either direction shall be not more than 0.0001 inch.

3.3.13.5 Absolute/incremental input. The control shall have absolute/incremental input capability. This mode shall be selected by a preparatory G-code or setpoint initialization switch switched on during the part program.

3.3.13.6 Automatic tape code recognition. The control shall automatically recognize and read tape punched either with EIA Standard RS-244 or EIA Standard RS-358 (ASCII) character codes.

3.3.13.7 Switchable inch/metric input. The control system shall provide switchable inch/metric capability.

3.3.13.8 Part program storage. Unless otherwise specified (see 6.2.1), part program storage shall be in either solid-state or bubble-type memories. Unless otherwise specified (see 6.2.1), the MCU shall store in

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memory the content of not less than 250 feet of postprocessed part program tape data.

3.3.13.9 Standby battery power. Unless otherwise specified (see 6.2.1), standby battery power shall be provided for controls having volatile memory. The standby battery power shall retain a fully loaded memory for a period of not less than 72 hours after the primary power source fails.

3.3.13.10 Part program edit. A program edit capability shall be provided to allow block insertion, deletion, or modification of any part program in the part program memory. Data related to part program edit shall be input manually. Edited program data entered into program memory shall be automatically acted upon, in the sequence designated, whenever the part program is cycled.

3.3.13.11 Feed rate override. A feed rate override control for each controlled axis shall be provided to allow manual adjustment of the programmed feed rate from zero to not less than 120 percent of the programmed rate in either continuous adjustment or in steps of not more than 10 percent. The feed rate override control shall have a position that allows return of feed rate to that originally programmed.

* 3.3.13.12 Spindle speed override. A spindle speed override shall be provided to allow manual adjustment of spindle speed in either continuous adjustment or in steps of not greater than 10 percent each. The spindle speed override control shall control not less than the range of override percentages between 60 and 120 percent of programmed speed. The spindle speed override control shall have a position that allows return of speed to that programmed.

3.3.13.13 Sequence number search. The control shall permit the operator to search for any desired sequence number. This search may be forward or backward. A sequence number readout display shall be provided on the control panel.

3.3.13.14 Preparatory functions (G-Codes). The preparatory function codes of the control shall be in accordance with EIA Standard RS-274.

3.3.13.15 Miscellaneous functions (M-Codes). The miscellaneous function codes of the control shall be in accordance with EIA Standard RS-274.

3.3.13.16 Block delete. The control shall have a block delete feature to permit by-passing of blocks of programmed data.

3.3.13.17 Buffer storage. The control shall have buffer storage for transferring command data from the tape reader to internal storage without delaying the next incoming command and without interrupting the machine functions. The internal storage shall store multiple blocks of data until required by the controlling devices.

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3.3.13.18 Block-by-block read. The control shall have a block-by-block read function for reading information one block at a time.

3.3.13.19 Automatic cycle. Unless otherwise specified (see 6.2.1), the control shall run an entire part program in a continuous cycle in the automatic mode. Once programmed, the control shall automatically run for an infinite number of parts. It shall be possible to interrupt the automatic cycle without loss of information.

3.3.13.20 Reversal error compensation. Reversal error compensation shall be provided for each axis of motion to compensate for drive gear backlash when slide reversal occurs. A means shall be provided to input new compensation values when required.

3.3.13.21 Axis calibration. Error compensation shall be provided to correct repeatable errors determined to exist due to leadscrew and feedback inaccuracies. A means shall be provided to input new error compensation values as required.

3.3.13.22 Additional features. Unless otherwise specified (see 6.2.1), the following additional numerical control features shall be provided:

a. Incremental and continuous jog for each axis.
b. Leading zero suppression or decimal point programming.
c. Inches per minute (ipm)/millimeters per minute (mmpm) programming in all modes of operation.
d. Edit keylock.

* 3.3.13.23 Operator control panel. Unless otherwise specified (see 6.2.1), the control unit shall have a control panel which provides at least the following control functions: Emergency Stop; Cycle Start; Optional Stop; Mode Selection; Feed Hold; Feedrate Override; Spindle Speed Override; Jog; Jog Direction; and EOB Stop. Indicator lights or the data display unit shall indicate the following: read errors, over-temperature, program stop, and end-of-program. All operating control switches or levers shall be convenient to the operator in his normal operating position.

3.3.13.23.1 Pendant or stanchion control. When specified (see 6.2.1), a pendant or stanchion control shall be furnished in lieu of an operator control panel. The pendant control shall be connected to the machine to enable the operator to control machine functions while in close observation position. Unless otherwise specified (see 6.2.1), the pendant or stanchion control shall be capable of the same functions as stated for the operator control panel in paragraph 3.3.13.23.

3.3.13.24 Reference zero. The control unit shall have a reference zero to move each axis to a reference limit switch to establish synchronization and reset all registers to zero.

3.3.13.25 Set zero. The control unit shall have a "set zero" feature to permit the specification axis register to be reset to zero establishing present axis location as zero.

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3.3.13.26 Program tryout. The control shall have a program tryout feature to permit new part programs to be read through the control without machine motions.

3.3.13.27 Operational software. Operational software shall be resident in the control and shall contain all logic necessary to operate the machine tool. The software shall provide for meaningful diagnostics relating to the utilization of the control, machine tool, part program, and operator's actions. The MCU executive interface and diagnostic programs from those controls having volatile memory shall be provided in the form of one-inch, eight-channel punched tape complying with EIA Standard RS-227 or other input form as appropriate for the input media specified (see 6.2.1).

3.3.13.28 On-line system diagnostics. The MCU shall perform a constant monitoring function of various activities and conditions of the control and machine. The diagnostic system shall communicate appropriate fault messages to the operator by means of the CRT or other display. Unless otherwise specified (see 6.2.1), the on-line diagnostic capability shall determine the following fault conditions and display appropriate error messages:

a. Axis feedback loop fail
b. Illegal program entry
c. Axis travel limits exceeded
d. Axis servo motor failure
e. Reader read error (when applicable)
f. Control console overheat
g. Axis servo drive on/off
h. Lubrication fault
i. Guard door open
j. Program error
k. Feed limit

3.3.13.28.1 Maintenance diagnostics system. Maintenance diagnostics software shall be furnished as nonvolatile memory, punched tape, or as specified (see 6.2.1), to perform for diagnostic checking of the MCU. The diagnostic software shall test, exercise, and display failures, at least to board level.

* 3.3.13.29 Peripheral equipment interface. When specified (see 6.2.1), the control shall be equipped with an interface complying with EIA Standard RS-232 for peripheral equipment.

3.3.13.30 Axis jog. Manual controls shall be provided to accomplish both single and continuous, low- and high-speed jog movements of the slides for each controlled axis in both the plus and minus directions.

3.3.13.31 Auxiliary functions. Unless otherwise specified (see 6.2.1), the MCU shall respond to the following program-commanded auxiliary function: interpolation, fixed cycles, and dwell. Unless otherwise specified

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(see 6.2.1), miscellaneous functions shall include program stop, optional program stop, end-of-program, spindle off, coolant on-off, end-of-program (rewind), and program re-start. System preparatory and miscellaneous functions shall be coded in accordance with EIA Standard RS-274.

3.3.13.32 Canned cycles. Canned cycles shall be provided as specified (see 6.2.1).

3.3.13.33 Tape punch unit. When specified (see 6.2.1), the system shall have a tape punch unit for producing punched tapes from edited programs of input tape and from programs stored in data memory. The unit shall be compatible with the tape punch output of the system, punching both paper and mylar tapes of dimensions, characteristics, and code in accordance with 3.3.13.1. The punching speed of the unit shall be not less than 75 characters per second.

3.3.13.34 Postprocessor. When specified (see 6.2.1), the system shall be furnished with a postprocessor program(s) to be utilized in off-line computer assisted preparation of part programs. The postprocessor shall convert specified tape information (see 6.2.1) to the form and format required to operate the machine tool and control. The postprocessor shall be written for use on the type computer specified (see 6.2.1).

3.3.13.35 Wheel wear compensation. When specified (see 6.2.1), the control unit shall compensate for new grinding wheels whose initial dimensions vary from nominally programmed values. The control shall compensate for different wheel diameters and widths by storing the difference between the programmed wheel size and its actual size (offset values) in a wheel diameter and wheel width offset table as entered by the operator.

3.3.14 Drive mechanisms. Means shall be provided to position each moving member in all axes of motion. Manual positioning shall be accomplished utilizing handwheels and precision leadscrews. Numerically controlled movement shall be accomplished by solid-state servo drives, electric motors, position and velocity feedback devices, and precision leadscrews or precision, preloaded, antifriction recirculating ball screw and nut assemblies forming a closed loop servo system. The leadscrews shall be hardened and precision ground to provide alignment and positioning accuracy, and be of a size, class, and precision suitable for the application. Leadscrews and nuts shall not exert excessive loads or cause premature deterioration of the feed drive motors or feed drive components. Feeds and positions are to be monitored by a feedback system and transmitted to the numerical control. Means shall be provided on all machines for proper lubrication. All slides and screws shall be shielded from contaminants by covers or guards.

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3.3.15 Position feedback device. A position feedback device including all related components, hardware and electrical equipment shall be provided for each axis having a readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device during operation of the machine. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis (see 3.3.13).

3.3.16 Hour meter. When specified (see 6.2.1), the machine shall be equipped with an hour meter. The hour meter shall display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type with a range of 0 to 99,999 hours in increments of 1 hour or less. The meter shall be sealed and mounted to withstand shock and vibration generated by the machine.

3.3.17 Standard equipment. Unless otherwise specified (see 6.2.1), the following equipment shall be furnished with each machine:

a. One set of special wrenches, operating, and repair tools that are normally furnished by the manufacturer.

b. One adjustable worklight to illuminate the work area, complete with protective shield.

c. Grinding wheels: manufacturer's standard set (not less than 3 wheels).

d. Tooling for machine calibrations and setup.

e. Dial indicator and gage block set.

f. One extra set of wheelhead belts, if applicable.

g. Work enclosure or splash guards to enclose work area.

3.3.18 Optional equipment. The following equipment and accessories are commercially available and are considered optional. When required, the quantity of each item and its characteristics shall be as specified (see 6.2.1):

a. Mounting devices.

b. Toolholder kits.

c. Variable speed spindles.

d. End mill adapters.

e. Grinding wheels: shank size, number of each wheel size, type, dimensional characteristics, and composition of each wheel shall be as specified (see 6.2.1).

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f. Pre-set tool setter.

g. End mill blanks.

3.4 Size and capacities. Unless otherwise specified (see 6.2.1), the size and characteristics of the machine shall be not less than those specified in table I for the machine size specified.

* 3.5 Performance. The grinder and numerical control system combination shall be capable of grinding workpieces in accordance with the requirements herein when submitted to the operational test given in 4.7.2 and the performance tests given in 4.7.3. Finished workpiece physical dimensions shall conform to ANSI B94.19. Diameter of cutter tooth edges, when measured at any point along their full length, shall be within -0.000 to +0.003 inch. Runout shall not exceed 0.002 inch total indicator reading (TIR). Surface finish shall be not rougher than 24 microinch.

3.6 Alignment and accuracy tolerances. Each machine shall meet the alignment and accuracy tolerances specified in table II when tested according to the procedures given in 4.7.1.

3.7 Repair parts. Such repair parts as are specified (see 6.2.1), shall be furnished.

3.8 Marking of instruction plates, indicating plates and nameplates. All words on instruction and indicating plates shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background. All plates shall be corrosion-resistant.

3.8.1 Lubrication plate. Unless otherwise specified (see 6.2.1), a lubrication plate shall be permanently and securely attached to each machine. The plate shall contain the following information:

Points of lubricant application
Servicing interval
Type of lubricant
Viscosity

3.8.2 Nameplate. A nameplate shall be permanently attached to each machine. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the machine is a special model, the model designation shall include the model of the basic standard machine and a suffix identified in the manufacturer's permanent records. The captions listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation

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Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.9 Technical data. Technical data (operating manuals, maintenance manuals, parts catalogs, prints, wiring diagrams, lubrication charts, and machine alignment and accuracy test results), when required, shall be furnished in accordance with the requirements of DD Form 1423. All technical data shall be written in the English language.

3.10 Workmanship. Workmanship of the machine shall be commensurate with all requirements specified herein and shall be of a quality equal to that prevailing among manufacturers producing equipment of the type covered by this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Quality conformance inspection (see 4.4).

b. Acceptance test (see 4.5).

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* 4.3 Inspection conditions. Unless otherwise specified (see 6.2.1), all quality conformance inspections, tests, and examinations (see 4.4) shall be performed in the manufacturer's designated indoor test area under normal environmental conditions and all acceptance test requirements (see 4.5) shall be performed only after the grinder is installed at its final location.

* 4.4 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of at least the examination in 4.6, the tests in 4.7.2 and 4.7.4, the accuracy and alignment requirements in 4.7.1, and the inspection in 4.8. Failure to pass the examination or test shall be cause for rejection.

* 4.5 Acceptance test. An acceptance test shall be performed on the grinder to ensure conformance with this specification. Unless otherwise specified (see 6.2.1), the acceptance test shall shall consist of the examination in 4.6, the accuracy and alignment requirements in 4.7.1, and all tests in 4.7.2 through 4.7.3.2.3. Failure of the grinder to pass one or more tests or any part of a test will be cause for rejection.

4.6 Examination. The grinder and equipment shall be examined for design, construction, material, components, electrical equipment, and workmanship to determine compliance with all requirements of this specification, including compliance with all referenced standards.

4.7 Tests. The machine shall be subjected to the following tests. All instruments, tapes, materials, and tools required to perform and evaluate these tests shall be furnished by the supplier. The measuring instruments shall have evidence of calibration traceable to the National Bureau of Standards. The numerical control test tapes and computer printouts used shall become the property of the Government.

4.7.1 Accuracy and alignment requirements. Each machine and its components shall be checked for compliance with the accuracies specified in table II. The instruments and methods used in performing the tests shall be at the option of the supplier provided they accurately determine the tolerances of the machine.

4.7.2 Operational test. The machine shall be operated at no load for not less than 1 hour. The machine and the numerical control system shall be operated in accordance with the manufacturer's standard operating test procedure for warm-up and run-off checks. During the warm-up period, proper operation of all controls, motors, adjustment mechanisms, and accessories shall be verified. After initial warm-up for a period of not less than 4 hours, the machine shall be cycled continuously under numerical control to verify its ability to receive and perform all program functions without error. This operation shall include spindle speed and feed rate changes that include the highest, intermediate, and lowest settings of each

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range, rapid traverse of all slides, simultaneous movement of slides, and automatic feed cycles, as applicable. The numerical control system shall be further tested to verify proper operation of MDI, program edit, and system diagnostics. Should a malfunction occur, it shall be corrected and the operational test repeated until a full 4 hours of running time is completed without failure. At completion of this test, the machine shall return to set-point within the allowable tolerance given in 3.3.13.

* 4.7.3 Performance test. Each grinding machine shall be tested as follows to ensure that its performance complies with the requirements of this specification. At the option of the procuring agency, each machine shall be subjected to grinding tests in accordance with specific requirements designated by the activity in lieu of or in addition to the following tests (see 6.2.1).

4.7.3.1 Wheel dressing and truing test. When wheel dressing and truing cycles are numerically controlled (see 3.3.6), they shall be tested before performing the grinding tests in 4.7.2.2 and 4.7.2.3. The methods used in performing this test shall be the manufacturer's option provided they accurately determine the ability of the components and control unit to properly dress and true the grinding wheel.

* 4.7.3.2 Grinding test. The grinding tests shall meet the following requirements. The tests shall be performed on a heat treated, high speed steel, fluted blank meeting the requirements for a 1-1/4 inch diameter, 4 flute, 3 inch flute length, 0.120 corner radius end mill. The blank shall meet the requirements of ANSI B94.19 except sufficient stock shall be left on the diameter and radial rake face to permit finish grinding of these surfaces. End geometry shall be omitted. Shanks shall be finish ground with Weldon Flats. Grinding wheel, wheel speed, feed rate, and depth and width of cut shall be manufacturer's option.

* 4.7.3.2.1 Flute grinding test. Grind the radial rake face on five end mill blanks (see 4.7.3.2). All five end mills shall be produced consecutively and from a single tape (program). Finished radial rake dimensions shall meet the requirements of 3.5.

* 4.7.3.2.2 End gashing test. Grind (gash) the end geometry (square end, center cutting) of the five end mill blanks (see 4.7.3.2). All five end mills shall meet the dimensional accuracies given in 3.5.

* 4.7.3.2.3 Relief and clearance grinding test. Grind the relief and clearance on the outside diameter, corner radius and end (face) on five end mill blanks (see 4.7.3.2). All five end mills shall meet the dimensional requirements given in 3.5.

4.7.4 Electromagnetic interference control tests. Equipment requiring electromagnetic interference control testing shall be tested in accordance with MIL-STD-461.

4.8 Packaging inspection. Packaging shall be inspected to determine compliance with the requirements of section 5.

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5. PACKAGING

5.1 Preservation, packing and marking. Unless otherwise specified (see 6.2.1), preservation, packing, and marking shall be in accordance with ASTM D 3951. When specified (see 6.2.1), level A preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-M-18058.

6. NOTES

6.1 Intended use. The numerically controlled tool and cutter grinders covered by this specification are intended to grind and resharpen conventional cutters within the range specified, including all types of milling cutters, reamers, counterbores, hobs, or other metal-cutting tools. They can be used for grinding and resharpening conventional type end milling cutters having cylindrical or tapered sides, straight or helical flutes, ball, radius, or square nose, and right- or left-hand spirals.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Specify type, style, and size required (see 1.2).

c. If the machine is required to be configured in a specific measurement system (US or SI), state required system (see 3.1.1).

d. If measuring and indicating devices are required to be graduated in specific measurement system (US or SI), state required system (see 3.1.1.1).

e. Specify additional safety and health requirements, if different (see 3.1.4).

f. Painting, if different (see 3.2.5).

g. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.2.10).

h. Tailstock taper or other internal configuration, if applicable (see 3.3.2.2).

i. Contouring or positioning rotary table, if required (see 3.3.2.3).

j. Special collet sizes, if required (see 3.3.3).

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k. Infinitely variable wheelhead speed range, if required (see 3.3.5).

l. Specify either one industrial diamond or a cluster diamond to dress and true the grinding wheel; programmable numerically controlled wheel truing and dressing cycle(s), if required (see 3.3.6).

m. Locator fixture, if required (see 3.3.8).

n. A coolant system, if not required (see 3.3.10).

o. Dust extractor, if required (see 3.3.11).

p. Electrical system, electrical supply, and voltage for initial wiring, if different (see 3.3.12).

q. Motor requirements, if different (see 3.3.12.1).

r. Specify axes to be numerically controlled; executive diagnostic and interface program routine format, if different (see 3.3.13).

s. Data input media for control, if different; specify EIA RS-358 or EIA RS-244; tape reader reels, if different (see 3.3.13.1).

t. Manual data input unit, if not required (see 3.3.13.1.1).

u. Data display, if different (see 3.3.13.2).

v. Part program storage memory type, if different; part program storage capacity, if different (see 3.3.13.8).

w. Standby battery power required, if different (see 3.3.13.9).

x. Automatic cycle, if not required (see 3.3.13.19).

y. Additional CNC features, if different (see 3.3.13.22).

z. Operator control panel functions, if different (see 3.3.13.23).

aa. Pendant or stanchion, as specified, if required; pendant or stanchion control panel functions, if different (see 3.3.13.23.1).

bb. MCU program input and storage media, if different (see 3.3.13.27).

cc. On-line diagnostic fault conditions, if different (see 3.3.13.28).

dd. Maintenance diagnostics system input form, as specified (see 3.3.13.28.1).

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ee. Peripheral equipment interface, if required (see 3.3.13.29).

ff. Auxiliary and miscellaneous functions, if different (see 3.3.13.31).

gg. Canned cycles, if required (see 3.3.13.32).

hh. Tape punch unit, if required (see 3.3.13.33).

ii. Postprocessor, if required; specify computer name, model, form, and format (see 3.3.13.34).

jj. Wheel wear compensation, if required (see 3.3.13.35).

kk. Hour meter, if required (see 3.3.16).

ll. Standard equipment, if different (see 3.3.17).

mm. Specify optional equipment, if required (see 3.3.18).

nn. Size, number, type, dimensional characteristics, and composition of each grinding wheel (see 3.3.18.e).

oo. Size and capacities, if different (see 3.4).

pp. Repair parts, if required (see 3.7).

qq. Lubrication plate, if different (see 3.8.1).

rr. Nameplate, if different (see 3.8.2).

ss. Inspection conditions, if different (see 4.3).

tt. Quality conformance inspection, if different (see 4.4).

uu. Acceptance test, if different (see 4.5).

vv. Performance tests, if different (see 4.7.3).

ww. If preservation, packing and marking in accordance with MIL-M-18058 are required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. All data requirements (parts lists, wiring diagrams, operating manuals, maintenance manuals, and alignment and test results) requested on a DD Form 1423, Contract Data Requirements List, shall be furnished in accordance with each specified DD Form 1664, Data Item Description, as shown on the DD Form 1423.

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* 6.3 Subject term (key word) listing.

Computer numerical control (CNC), grinder
Dressing and truing, wheel
Grinder, tool, cutter, flute, radius
Machine tool, grinding, tool, cutter, flute, radius
Metalcutting tools, counterbores, hobs, milling cutters, reamers
Tailstock
Wheel, grinding, abrasive
Wheelhead, horizontal spindle
Workhead rotary table assembly

* 6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements in this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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* TABLE I. Sizes and capacities.

Characteristics	Type I			
	Style A		Style B	
	Size 1	Size 2	Size 1	Size 2
Capacity:				
Number of NC axes	← 2 to 4 →		3 or 4	3 to 6
Longitudinal table travel (inches)	18-1/2	24	17-1/2	24
Cross slide travel (inches)	9	9	10-1/4	12
Swing over table (inches)	10	10	11	12
Distance between centers of workhead and tailstock (inches)	26	33	17-1/2	35
Table working surface (inches)	31 x 4	52 x 6	31 x 4	52 x 6
Swivel table horizontal swivel (degrees)	200	360	--	--
Maximum grinding diameter (inches)	10	10	11-1/2	11
Workhead:				
Horizontal swivel (degrees)	← 360 →			
Tilts (degrees)	±30	±30	±30	360
Wheelhead:				
Number of spindle speeds	← 4 →			
Vertical travel (inches)	11-1/2	11-1/2	9-1/2	12
Horizontal swivel (degrees)	← 350 →			
Tilt (degrees)	← -50 to +25 →		← ±90 →	
(RPM)	← 2,250 to 6,000 →			
Grinding motor horsepower (hp)	2	2	2	3

NOTE 1: Machine size and capacities shall be not less than the stated requirements of the size ordered shown in above table. When the largest size and capacity machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

NOTE 2: Dashes (--) indicate a feature not available on the size machine listed.

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* TABLE I. Sizes and capacities - continued.

Characteristics	Type II	
	Size 1	Size 2
Capacity:		
Number of NC axes	5	7
Maximum end mill shank diameter (inches)	2-1/2	2-1/2
Minimum end mill shank diameter (inches)	1/4	1/4
Maximum flute length (inches)	8	10
Maximum overall length of tool (inches)	10	16
Maximum cutter diameter (arbor-style) (inches)	6	10
Maximum grinding radius (inches)	--	4
Maximum flute length-with ball nose	--	8
-with 2" diameter or larger	--	8
Axes travel:		
Wheelhead:		
Cross axis (inches)	12	18
Vertical (inches)	18	18
Workhead:		
Longitudinal base slide (inches)	18	24
Saddle (inches)	8 (manual)	12
Table (inches)	4 (manual)	8
Spindle rotation (degrees)	360	360
Rotary table swivel (degrees)	205	220
Wheelhead:		
Wheel spindle rotation (RPM) (standard)	6000	4500
Variable speed (RPM) (optional)	720 to 7200	1000 to 6000
Standard wheel diameter (inches)	6	8
Grinding wheel motor horsepower (HP)	5	10
Feedrates:		
Wheelhead:		
Cross axis (IPM)	0-200	0-200
Vertical (IPM)	0-200	0-150
Workhead:		
Longitudinal (IPM)	0-200	0-200
Saddle (IPM)	(manual)	0-200
Table (IPM)	(manual)	0-200
Spindle rotation (degrees/min)	54,000	36,000
Rotary table swivel (degrees/min)	1,200	700

NOTE 1: Machine size and capacities shall be not less than the stated requirements of the size ordered shown in above table. When the largest size and capacity machine shown in above table is ordered, the size and capacity of the machine offered shall not exceed the stated requirements by more than 10 percent. When a range is shown, the required performance is from the stated minimum or less to the stated maximum or greater.

NOTE 2: Dashes (--) indicate a feature not available on the size machine listed.

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* TABLE II. Alignment and accuracy requirements.

Component characteristics	Permissible tolerance
For type I and type II machines:	
Worktable assembly:	
Flatness of work mounting surface	0.001"/FT
Squareness of work mounting surface to spindle axis, in any 6" sweep radius (0° to 180° method)	0.0005"/FT
Longitudinal motion, rise and fall, with respect to the spindle	0.0012"/FT
Cross travel, rise and fall, with respect to the spindle	0.0012"/FT
Parallelism of T-slots with longtudinal movement	0.0005"/FT
Squareness of T-slots with cross axis movement	0.0005"/FT
Workhead spindle runout	0.0002" TIR
Grinding spindle:	
Wheel spindle end play and runout	0.0002" TIR
Squareness of spindle movement to the work mounting surface per 6" of travel	0.00025"/6"
Concentricity of the wheel spindle:	
-straight diameter	0.001" TIR
-face	0.001" TIR
Positioning accuracy and repeatability:	
Linear axes positioning and repeatability accuracy:	
-absolute positioning	1/ ±0.001"/FT
-positioning repeatability	±0.0003"
Rotary axes (when furnished) positioning and repeatability accuracy:	
-absolute positioning	±20 sec
-positioning repeatability	±10 sec
Lead screw reversal error	0.0002 inch
For type I machines:	
Tailstock:	
Tailstock spindle parallel with table	0.0012"/FT
Tailstock axis in line with workhead axis:	
- in vertical plane	0.0008" TIR
- in horizontal plane	0.0008" TIR
For type II machines:	
Rotary table parallel to base slide in the horizontal plane over 90° of travel	0.001" TIR

1/ Not to exceed 0.025" in overall travel.

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Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - ME
Air Force - 84
DLA - GS
Navy - No Interest

User activities:
Army - SM, AR
Navy - MC

Preparing activity:
DLA-IP

Project Number: (3415-0161)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-80266A	Grinding Machines, Tool and Cutter, Flute and Radius, Computer Numerically Controlled (CNC)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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